DOMAIN 1 · FOUNDATIONS

Foundations of Accounting

The one record behind every cost report, invoice and forecast — built from first principles, for the project-controls professional.

5 Knowledge Areas   **~55 min** lecture   Certified Project Controls Professional — **AI**

↓ Press → or **Space** to begin ↓

HOW THIS LECTURE WORKS

Paced by you, one idea at a time

Built the way the evidence says people actually learn from lecture video — short segments, revealed step by step, each concept paired with a picture, and a quick check to make it stick.

- → **Advance a step** — → or Space. Go back with ←. Each slide builds; take your time.
- ≣ **Speaker notes** — press N for the narration behind every slide (record it, or read along).
- ◐ **Light or dark** — press T. The whole deck is designed for both.
- ✓ **Knowledge checks** close each Knowledge Area — answer before you move on.

WHY THIS MATTERS TO PROJECT CONTROLS

Every number you report is a view into one record

Cost reports, EV curves, invoice applications, forecasts — all of them draw, ultimately, on the same double-entry ledger. Domain 1 builds that record so the figures stop being a black box.

KA 1.1

The accounting model

Equation, debits & credits, journal → ledger → trial balance.

KA 1.2

The statements

The four financial statements and how they lock together.

KA 1.3

Accrual & matching

Recording economic events, not cash movements.

KA 1.4

Provisions & accruals

Cost you know is coming — the IAS 37 judgement.

KA 1.5

Chart of accounts

Cost coding, the CBS and mapping to the WBS.

1.1

The accounting model

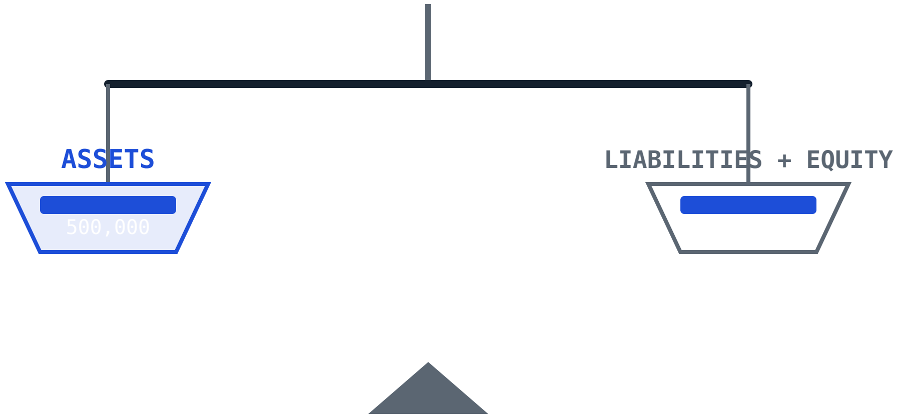
By the end you will be able to:

1. State the accounting equation and explain why every transaction keeps it in balance.
2. Apply debit and credit rules across the five account types.
3. Follow one transaction cycle: journal → ledger → trial balance → equation.
4. Explain what a trial balance proves — and, crucially, what it cannot.

KA 1.1 · THE ACCOUNTING EQUATION

Assets = Liabilities + Equity

The identity underlying the IFRS Conceptual Framework — true for a company, a joint venture, or a single-project vehicle. It holds because of **double aspect**: every event is recorded twice, once on each side of the scale.



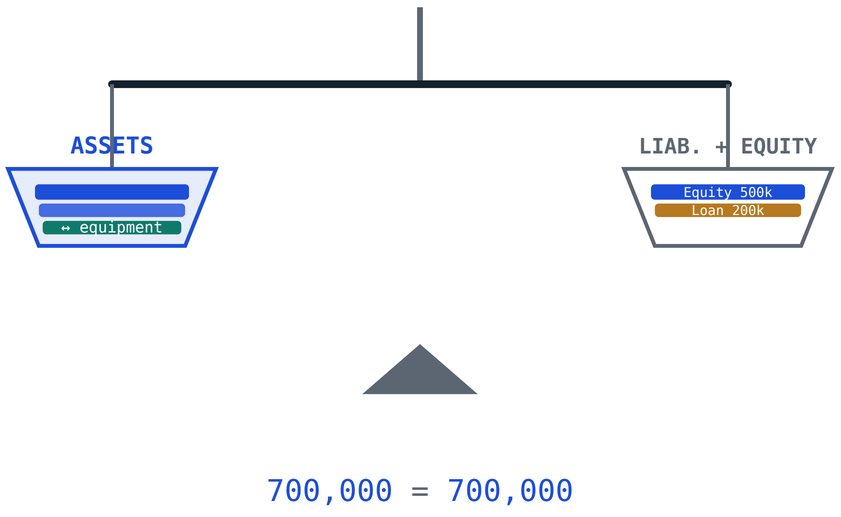
Double aspect. Borrow cash → an asset *and* a liability rise together. Buy kit for cash → one asset falls, another rises. The scale never tips.

KA 1.1 · WORKED EXAMPLE — MERIDIAN RAIL CONTRACTING

Three transactions, one level scale

DAY-ONE EVENTS	USD
1 · Share capital paid in cash	500,000
2 · Bank loan drawn in cash	200,000
3 · Site equipment bought for cash	150,000
Total assets	**700,000**

The classic trap. Transaction 3 swaps cash for equipment — assets' *composition* changes, the *total* doesn't. Not every entry hits profit.



KA 1.1 · DEBITS & CREDITS

Five account types, one rule each

Debit (Dr, left) and **credit** (Cr, right) carry no moral meaning — they're a labelling system. What matters is each account type's **normal balance**: the side on which it increases.

ACCOUNT TYPE	INCREASES WITH	NORMAL BALANCE	EXAMPLES
Asset	Debit	Debit	Cash, receivables, equipment
Expense	Debit	Debit	Labour, materials, depreciation
Liability	Credit	Credit	Loans, payables, provisions
Equity	Credit	Credit	Share capital, retained earnings
Income	Credit	Credit	Contract revenue, interest income

Hold it as one idea: uses of resources (assets, expenses) are debit-normal; sources (liabilities, equity, income) are credit-normal. And every entry obeys $\Sigma\ \mathrm{Dr} = \Sigma\ \mathrm{Cr}$.

KA 1.1 · THE CYCLE

Journal → ledger → trial balance → equation

Four steps turn raw events into checked figures. Capture first, classify second, prove third, close fourth — the same mechanism for a single-project vehicle or a multinational.

Journal	→	Ledger	→	Trial balance	→	Equation
every event, in date order		accounts collect Dr / Cr		Σ Dr = Σ Cr		A = L + E holds

> Meridian records **11 transactions** in March 2026 — capital, a loan, equipment, materials, wages, a subcontractor, a client invoice, a receipt, a supplier payment and depreciation. Watch one account fill.

KA 1.1 · POSTING TO THE LEDGER

The Cash T-account, filling

Cash (asset)

DEBIT · IN		CREDIT · OUT	
Share capital	500,000	Equipment	150,000
Loan drawn	200,000	Wages	45,000
Client receipt	150,000	Supplier paid	50,000

850,000 − 245,000 = closing balance **605,000 Dr**

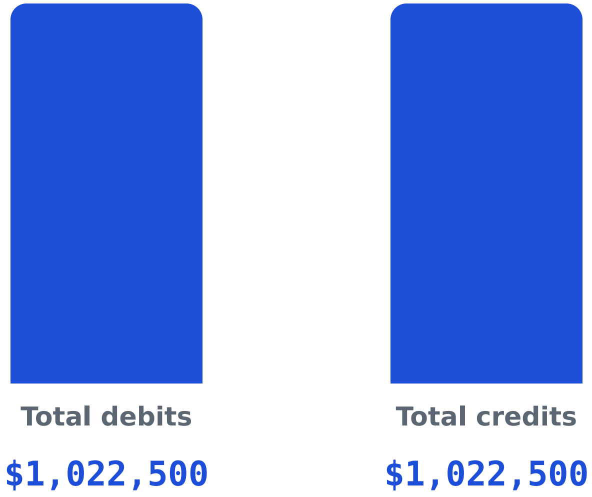
Posting is **purely mechanical** once the entry is classified: each line lands in its named account, tagged with a reference back to the journal.

> **Control accounts.** Where volume is high, a subsidiary ledger holds supplier-level detail and a single control account carries the total — the two must reconcile every period.

KA 1.1 · THE TRIAL BALANCE

Both columns prove equal

Every ledger balance, split Dr / Cr. If the two totals agree, the double-entry was applied consistently across the whole ledger.



Equal, to the dollar — the books are **in balance**.

KA 1.1 · THE LIMIT — READ THIS TWICE

What a balanced trial balance cannot catch

It proves only that $\Sigma\ Dr = \Sigma\ Cr$. Any error that preserves that identity slips straight through.

Error of principle

Right amount, wrong account of the *same* normal side (materials → labour). Totals unchanged.

Omission

A transaction never recorded at all. Both sides simply lower together.

Commission

Posted to the wrong account of the same type — say, the wrong supplier.

Compensating

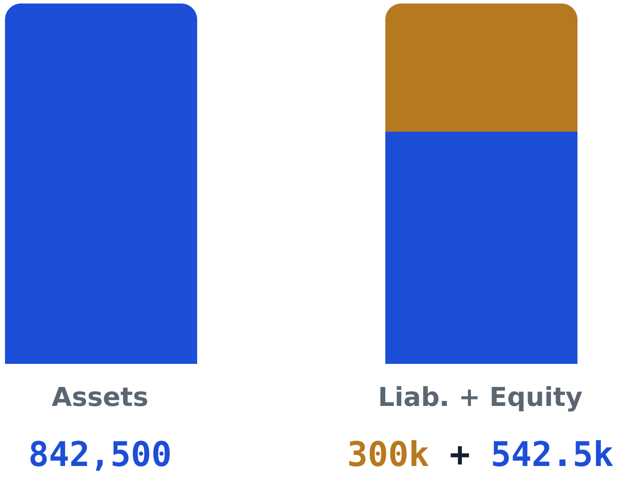
Two unrelated errors that happen to offset each other exactly.

Exam-critical: a balanced trial balance is **necessary but not sufficient**. In project controls these errors surface not from the trial balance but from cost-code and WBS reconciliation (KA 1.5).

KA 1.1 · CLOSING THE LOOP

Profit becomes equity; the equation holds

ROLL-UP, MARCH 2026	USD
Contract revenue	220,000
Less all costs	(177,500)
Net profit	**42,500**
Opening equity (capital)	500,000
Closing equity	**542,500**



842,500 = 300,000 + 542,500 ✓

KA 1.1 · KNOWLEDGE CHECK

A trial balance balances, yet a $15,000 subcontract cost was posted in full to “Labour cost.” Why did the trial balance *not* catch it?

- A The trial balance only checks liabilities, not expenses.
- B Both accounts are debit-normal expenses, so total debits were unchanged.
- C It caused an equal, opposite understatement of accounts payable.
- D A trial balance can’t be prepared when such an error exists.

B is correct. Moving an amount between two accounts on the *same* normal side leaves Σ Dr = Σ Cr untouched — an error of principle the trial balance is blind to. It’s found by account-level or cost-code review, not the trial balance.

1.2

Components of the financial statements

1. Name the four primary statements and the question each answers.
2. Read a Statement of Financial Position as the accounting equation, frozen in time.
3. Explain how the statements **articulate** — lock together into one coherent picture.

KA 1.2 · OVERVIEW

Four statements, four questions

IAS 1

Statement of Financial Position

What we **own and owe** at a point in time — the equation, frozen.

IAS 1

Profit or Loss & OCI

How we **performed** over the period — income less expenses.

IAS 7

Cash Flows

Where **cash** moved — operating, investing, financing. Profit ≠ cash.

IAS 1

Changes in Equity

How the **owners' stake** moved — profit, and distributions.

Plus the notes. Basis of preparation, accounting policies and the judgements behind the numbers — where the real story of a construction contract often lives.

KA 1.2 · ARTICULATION

They lock together

The statements share figures by design — this is why a set that "articulates" is internally consistent, and why an error in one surfaces in another.

Profit $42,500 → Retained earnings into equity → Equity $542,500 → Balance sheet A = L + E

> And separately: the **cash flow** statement's closing figure — Meridian's `$605,000` — *is* the cash line on the balance sheet. Same number, two windows.

1.3

Accrual accounting & matching

1. Contrast the cash basis with the accrual basis.
2. Apply the matching principle — cost recognised with the revenue it earns.
3. Post an accrual for work done but not yet invoiced, and reverse it.

KA 1.3 · TWO BASES

When do you record it?

CASH BASIS

When money moves

Record on receipt or payment. Simple — but it distorts performance: an unpaid subcontractor bill makes a bad month look good.

ACCRUAL BASIS · REQUIRED BY IFRS

When the event occurs

Record the cost when work is done and the revenue when it’s earned — whatever the cash is doing. The only basis that gives a true cost report.

The matching principle ties them: recognise a cost *in the same period* as the revenue it helped earn — not when its invoice happens to arrive.

KA 1.3 · WORKED EXAMPLE — THE ACCRUAL

Cost done, invoice not yet arrived

Month-end: Meridian’s subcontractor has completed $18,000 of verified work but has not invoiced. Cash basis records nothing — and understates the month.

> **Why controls care:** value of work done, not cash paid, is what earned value and the cost report must reflect. Missing accruals are the classic cause of a “great month” that reverses.

PERIOD-END ADJUSTING ENTRY

Dr Subcontract cost	18,000
Cr Accrued expenses	18,000

REVERSED ON DAY 1 NEXT PERIOD

Dr Accrued expenses	18,000
Cr Subcontract cost	18,000

…so the real invoice, when it lands, isn’t counted twice.

1.4

Cost provisions & accruals

1. Place an item on the certainty spectrum: payable → accrual → provision.
2. Apply the IAS 37 three-part recognition test.
3. Recognise a contract-loss (onerous-contract) provision — a real project risk.

KA 1.4 · THE CERTAINTY SPECTRUM

From invoice to informed estimate

The same idea — a cost the entity owes — at three levels of certainty. As certainty falls, judgement rises.

Trade payable
invoice received — amount & timing **certain**

›

Accrual
work done, not invoiced — amount **estimated**

›

Provision
uncertain **timing or amount** — IAS 37 judgement

> **IAS 37 defines a provision** as “a liability of uncertain timing or amount.” The question is: when are you *allowed* to recognise one?

KA 1.4 · IAS 37 RECOGNITION

Three gates — all must open

1

Present obligation

A legal or constructive duty **from a past event**. No past event → stop.

2

Probable outflow

Settlement will **probably** require resources (more likely than not).

3

Reliable estimate

The amount can be **reliably estimated**.

✓ **Recognise a provision**

Fail gate 2 → **contingent liability** (disclose in the notes, don’t book). Fail gate 1 → nothing at all.

KA 1.4 · TWO PROVISIONS THAT MATTER ON PROJECTS

Warranty, and the loss-making contract

WARRANTY PROVISION

Expected value, discounted

Post-handover rectification is probable and estimable. Measure at expected value; discount for timing → e.g. $95,238 today.

ONEROUS CONTRACT

Provide the whole loss, now

When unavoidable costs to finish **exceed** revenue left to earn, the full expected loss is provided for **immediately**. You can’t spread bad news.

AI in this topic. Models can flag a contract trending onerous from the cost forecast — a genuine early-warning use. But recognising the provision is a judgement the professional owns. *AI proposes; the professional disposes.*

1.5

Chart of accounts & cost coding

1. Read a project cost code and what each segment means.
2. Connect the cost breakdown structure to the WBS and the control account.
3. Judge where AI genuinely helps with coding — and where it must not decide.

KA 1.5 · ANATOMY OF A COST CODE

One code, many views

Each segment is an address. Together they let a single ledger line be sliced by work package, cost type *and* natural account.

07	.	30	.	03	.	530
WBS ELEMENT		**COST ELEMENT**		**SUB-ELEMENT**		**NATURAL ACCOUNT**
Pipeline Seg. 4B		Mechanical		Pipe-laying		Site labour

> **Control account = WBS × OBS.** Where a slice of work meets the team accountable for it, you get the control account — the cell where scope, cost and schedule are actually managed, and where cost coding, the CBS and the WBS all reconcile.

KA 1.5 · AI IN COST CODING

Where AI helps — and where it stops

GENUINELY USEFUL

Auto-classify & flag

Suggest a cost code from past patterns; surface anomalies — a labour code with no timesheet hours (the very error the trial balance missed).

THE GOVERNANCE LIMIT

It doesn’t know *why*

It learns past mistakes as readily as good practice, and can’t explain a break. Material or novel items route to a human.

The principle of the whole programme: *AI proposes; the professional disposes.*

DOMAIN 1 · THE THROUGH-LINE

One connected system

1.1 Record — equation, double-entry → 1.2 Report — four statements → 1.3 Timing — accrual & matching → 1.4 Judgement — provisions (IAS 37) → 1.5 Code — onto the WBS

> You don’t need to be an accountant. You need to read the record with confidence — and now you can trace any reported number back to where it came from.

END OF DOMAIN 1

“AI proposes; the professional disposes.”

Next — **Domain 2: Financial Reporting & the Standards.** The foundations you’ve just built meet IFRS in depth, including revenue recognition on construction contracts under IFRS 15.